

Relationships Between Angle Pairs

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Complementary Angles are two angles whose measures add up to 90° .

Supplementary Angles are two angles whose measures add up to 180° .

Adjacent Angles are angles that share a common vertex and a common side without overlapping.

Vertical Angles

Vertical Angles are opposite angles formed when two lines intersect.

Vertical angles are **congruent**.

Congruent Angles have equal measures.

Examples

Example 1

Two angles are complementary. One angle measures 35° . Find the other angle.

$$90^\circ - 35^\circ = 55^\circ$$

The other angle measures 55° .

Example 2

Two angles are supplementary. One angle measures 128° . Find the other angle.

$$180^\circ - 128^\circ = 52^\circ$$

The other angle measures 52° .

Example 3

Two vertical angles are formed by intersecting lines. One angle measures 67° .
Vertical angles are congruent.

$$\boxed{67^\circ}$$

The vertical angle measures 67° .

Practice Exercises

Practice 1

Angle A measures 25° . Angle B is complementary to Angle A .

Find the measure of Angle B .

Practice 2

Angle P measures 145° . Angle Q is supplementary to Angle P .

Find the measure of Angle Q .

Practice 3

Two vertical angles are formed by intersecting lines. One angle measures 67° .

Find the measure of its vertical angle.

Practice 4

Two adjacent angles form a straight angle.

One angle measures 72° .

Find the measure of the other angle.

Practice 5

Angle M and Angle N share a common vertex and a common side. Their interiors do not overlap.

What type of angle pair are they?

Activities

Activity 1

Two angles are complementary. One angle measures 58° .

Find the measure of the other angle.

Activity 2

Two angles are supplementary. One angle measures 143° .

Find the measure of the other angle.

Activity 3

Angle C measures 37° . Angle D is complementary to Angle C .

Find the measure of Angle D .

Activity 4

Angle E measures 125° . Angle F is supplementary to Angle E .

Find the measure of Angle F .

Activity 5

Two vertical angles are formed by intersecting lines. One angle measures 114° .

Find the measure of its vertical angle.

Activity 6

Two adjacent angles form a straight angle.
One angle measures 105° .
Find the measure of the other angle.

Activity 7

Angle G and Angle H share a common vertex and a common side. Their interiors do not overlap.

What type of angle pair are they?

Thank you!